



Oreo manual update

The 'Check for updates' button on Oreo now manually initiates an update from user end, regardless of prior scheduling. <http://dgit.in/DeOreo>



Anniversary ThinkPad

25th Anniversary ThinkPad design reportedly leaked, sports elements from original range. <http://dgit.in/ThinkOld>

little eerier because they can be vocal, instrumental, or both. Though it's rare to hear an individual's voice (such as the vocals on that Celine Dion song you hate), people have reportedly heard Christmas carols, songs from their youth or even entire symphonies. There's one common trait though – every individual who has experienced MH has heard a tone they're familiar with. There's no composition of music happening in these hallucinations.

WHO CAN EXPERIENCE IT?

Broadly speaking, individuals with hearing loss are most at risk. And even amongst the hearing impaired, the MH phenomenon is incredibly rare (<http://dgit.in/MHStudy1>). Because we don't know why it happens, we're not able to pinpoint the type of defect or the individuals who might be prone to it. According to the same study mentioned above, apart from the hearing impaired, it's also found in women, and among individuals who live alone. It's not a psychological defect either, because the people who hear it are perfectly mentally normal otherwise.

HOW DOES IT HAPPEN?

This is where it gets interesting. Numerous studies over the years have failed to pinpoint an exact cause for something as complex as hearing Beethoven's 9th symphony out of nowhere. Everything from epilepsy to intoxication has been found as a major underlying factor in a large number of individuals who have experienced musical hallucinations at some point in their lives. But trumping all of that, there could be a bigger cause behind the phenomenon – our own brain.

THE COMPOSER IN YOUR HEAD

The study (<http://dgit.in/MHStudy2>) by a group of researchers from the Institute of Neuroscience, Medical School, Newcastle University sheds light upon the influence that our brain has on these phantom melodies. Supported by the Wellcome Trust, the study begins by explaining the problems behind previously adopted methods of establishing

AUDIO ILLUSIONS

The mind isn't the only one that can play tricks on your ears. Check out our story on audio illusions used by sound engineers all the time here: <http://dgit.in/AudTrix>.

the brain as a source of musical hallucinations. Briefly, they were as such:

- Monitoring a subject across sessions does not take into account changes in brain activity that could be influenced by factors other than hallucination. For instance, what if the subject has just seen a horror movie and is obsessed with it – while not considering it significant enough to mention it to the researchers? These cinephiles, we tell you!
- Monitoring sessions across subjects does not take into account variations at a neuroanatomical level – simply put, two people with different expertise in music could have their brains wired differently to respond to it. For instance, a music conductor at an award winning opera might experience the hallucination music differently than an average joe with regular exposure to music.

That brings us to the approach that this study took – monitoring one individual across the same session. And they had some luck with that. The subject was a 60-something lady, referred to as Sylvia by the doctors, who had considerable expertise in music over the years. So much so, she could note down the musical notations of the tones she was hearing during her musical hallucinations.

THE METHOD

There is an established method that is applied to suppress tinnitus – listening to music or another sound that takes the hallucination away. Researchers for this study tried using the same for musical hallucination and

it worked, giving them the ability to switch the MH on and off in a single brain. Observing the brain activity on Sylvia over durations of silence versus durations of high intensity MH shed light not only on what causes the hallucinations, but also on how our brain perceives the world around us.

AND WHAT DOES IT TELL US?

The study revealed that the parts of the brain that we use for listening to music are the most active when Sylvia experienced the highest intensity of her MH. This supported a previous theory by Karl Friston that our brain essentially works as a prediction engine – using past experiences as a guide to prepare for what's coming next. In the case of music, the brain uses tones heard previously to predict what it should hear next.



Analysing the brain activity in a person suffering from MH reveals just how much the brain can alter our perception of reality

In an individual suffering from hearing loss, this prediction gradually supersedes their auditory perception of reality. This also explains why listening to actual music soothes the MH – it corrects the incorrect predictions being made by the brain during the experience.

Why does it only generate music, though? Perhaps because it is much more predictable (<http://dgit.in/BioMusic>), compared to a random combination of sounds out of nowhere. However, we're still pretty ignorant about this rather scary phenomenon. **4**